

## Common questions

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7/1      | (a) | (i)   |  | all points plotted correctly (2)<br>5/6 points plotted correctly (1)      tolerance $\pm\frac{1}{2}$ small square<br>suitable curve (1)                                                                                                                                                                                                                                                                                                   |                 | 3   |     | 3     | 3     |      |
|          |     | (ii)  |  | increase until 1980/1990/late 1980s<br>(allow specified year e.g.1987) (1)<br><br>decrease after 1980/1990/late 1980s<br>(allow specified year e.g.1987) (1)<br><br>award (1) only for simple statement referring to an initial increase<br>followed by a decrease                                                                                                                                                                        |                 |     | 2   | 2     |       |      |
|          |     | (iii) |  | award (1) for any of following <ul style="list-style-type: none"> <li>only one reading every 10 years</li> <li>10 years between every reading</li> <li>graph does not go up one year at a time</li> <li>emissions very similar in 1980 and 1990 and there is no way of knowing what happened in between</li> <li>there could have been a lag in the reduction of sulfur dioxide emissions after the regulation came into force</li> </ul> |                 |     | 1   | 1     |       |      |
|          | (b) |       |  | $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$<br><br>$\text{H}_2\text{O}$ (1)<br>balancing (1)<br><br>balancing mark only awarded if $\text{H}_2\text{O}$ correct                                                                                                                                                                                                                                       |                 | 2   |     | 2     | 1     |      |
|          |     |       |  | Question 7/1 total                                                                                                                                                                                                                                                                                                                                                                                                                        | 0               | 5   | 3   | 8     | 4     | 0    |